Identity and decision problem

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Verification code: AI-A2-7953 | Personal seed: 7953

Decision setting

A Musanze potato cooperative receives inconsistent farm spreadsheets. Its field officer needs an offline decision-support service that estimates harvest quantity, identifies farms likely to have below-median yield per hectare, and groups farms with similar input conditions.

Three machine-learning questions

1. Regression: how many tons might this farm harvest?
2. Classification: is the farm estimated to be in the high-risk class?
3. Clustering: which farms have similar measured input conditions?

Ethical boundary

This is a synthetic classroom prototype, not an agronomic authority. It must not replace an agronomist or determine access to finance, insurance, public services, or farm treatment. A human field officer remains responsible for decisions.

Data and reproducibility

The seed is int(last four registration digits) = 7953. Base rows = 240 + (7953 mod 41) = 280. Three exact duplicates produce 283 raw rows; cleaning returns 280 rows.

Field	Generation / meaning
rainfall_mm	Uniform 600–1400
avg_temp_c	Uniform 12–26
soil_moisture_pct	Uniform 20–80; 14 planned missing position
fertilizer_kg	Uniform 0–250; 14 planned missing position
field_size_ha	Uniform 0.2–4.0
season	Random A or B
cooperative_support	Random binary 0 or 1
yield_tons	Prescribed formula plus N(0, 0.80), clippe
high_risk	1 when base-row yield/ha is strictly below

Byte-level signature

CSV: data/AI_A2_2527953.csv

SHA-256: ffb9be9b7e281791dc2143236fbb19a067ff3bf5e793b535f90b2b280ed149d4

The same default_rng(7953), draw order, four-decimal formatting, and newline convention recreate the same bytes. The risk median is computed on the original base rows before missing values and duplicates.

Reproducible flow

Generate → validate raw schema/ranges/nulls/duplicates → remove exact duplicates → one stratified 80/20 split → training-only imputation/scaling/encoding → cross-validation on training rows → fit selected pipelines → held-out evaluation → save models, predictions, JSON and plots → CLI/API inference.

Inputs and transformations

Only rainfall_mm, avg_temp_c, soil_moisture_pct, fertilizer_kg, field_size_ha, season and cooperative_support are inputs. Numeric fields use median imputation and standardization. Season uses most-frequent imputation and one-hot encoding. Every transformer is inside a scikit-learn Pipeline and is fitted only on training data.

Leakage prevention

yield_tons, high_risk, yield per hectare, row_id and all target-derived values are excluded from classification and clustering inputs. Missing values are not globally filled. The test set is not used for model selection; five-fold training-only CV selects the final regression and classification estimators.

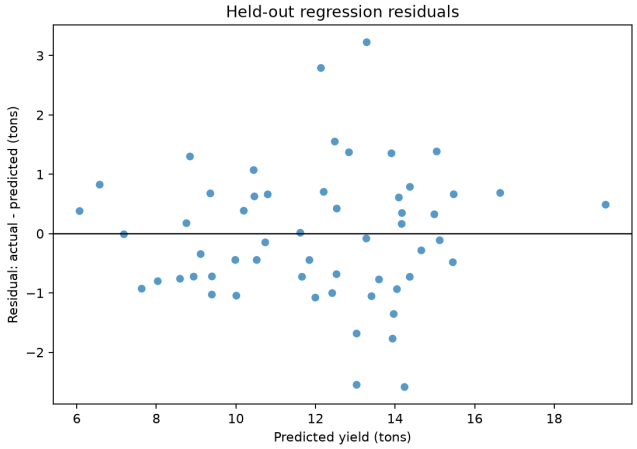
Task	Models compared	Selection evidence
Regression	Mean dummy; Ridge; Random Forest	Lowest CV MAE; RMSE tie-break
Classification	Majority dummy; Logistic; Random Forest	Highest CV F1; recall tie-break
Clustering	K-Means K=2,3,4,5,6	Highest training silhouette

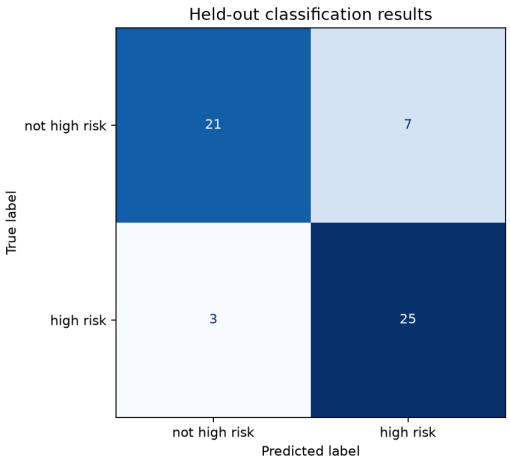
Regression model	MAE	RMSE	R²
dummy_mean	2.415	2.944	-0.000
random_forest	1.340	1.634	0.692
ridge	0.870	1.102	0.860

Ridge was selected from training-only CV (mean MAE 0.812). On test data it achieved MAE 0.870 tons and R² 0.860.

Classifier	Precision	Recall	F1	ROC-AUC
dummy_majority	0.000	0.000	0.000	0.500
logistic_regression	0.806	0.893	0.847	0.943
random_forest	0.781	0.893	0.833	0.917

Random Forest was selected because its training CV F1 (0.939) narrowly exceeded Logistic Regression (0.938). Held-out selection was not used.





Largest residual: row 150, actual 16.510 t, predicted 13.285 t (3.225 t under-estimate).

K	Silhouette	Training counts
2	0.1568	{'0': 110, '1': 114}
3	0.1293	{'0': 75, '1': 86, '2': 63}
4	0.1363	{'0': 54, '1': 70, '2': 55, '3': 45}
5	0.1340	{'0': 51, '1': 43, '2': 32, '3': 62, '4':
6	0.1409	{'0': 39, '1': 40, '2': 38, '3': 37, '4':

K=2 has the highest observed silhouette (0.1568). The value is modest, so the grouping should be treated as a useful description rather than evidence of sharply separated natural farm types.

ID	N	Rain mm	Fert kg	Field ha	Support
0	110	1021.2	128.6	2.21	0.00
1	114	1017.4	123.4	1.98	1.00

Measured interpretation for student review: the strongest difference is cooperative support—cluster 0 contains unsupported farms while cluster 1 contains supported farms. Other average conditions are fairly similar. IDs 0 and 1 are labels, not best/worst rankings.



Interface	Contract
GET /health	200 with status and AI-A2-7953; 503 if mod
GET /model-info	Identity, model names, K, features, finger
POST /predict	Seven typed fields; yield, risk, probabili
predict.py	Loads the same saved pipelines; never retr

Validation behavior

The API rejects missing fields, extra fields, strings pretending to be numbers, nulls, NaN/infinity, invalid seasons/support flags, and numeric values outside the prescribed ranges. FastAPI returns HTTP 422 rather than an unhandled server error. The invalid CLI exits nonzero with a readable validation message.

Executed test evidence

The suite covers valid inference, missing fields, wrong types, ranges, extra inputs, seed mismatch, byte reproducibility, exact duplicates, planned missingness, leakage allowlists, model loading, K=2…6 evidence, API responses, health/model information, and CLI/API parity. See evidence/AI_A2_2527953_TEST_LOG.pdf for the actual command and result.

Example valid record

{"rainfall_mm":950.0,"avg_temp_c":18.0,"soil_moisture_pct":55.0,"fertilizer_kg":90.0,"field_size_ha":1.5,"season":"A","cooperative_support":1}

Intended use and users

A classroom demonstration for a cooperative field officer: offline comparison of synthetic farm conditions, estimated yield, and a high-risk screening signal. Predictions should support questions, not settle decisions.

Risks and prohibited use

Synthetic-only training may omit real weather, disease, soil chemistry, management and market effects. Errors could misdirect scarce attention. Probability estimates are not calibrated agronomic truths. Outputs must never determine credit, insurance, public-service eligibility or treatment without consent, real validation and qualified review.

Measured errors requiring Kawaya's review

Regression: row 150 was under-estimated by 3.225 tons; relying on that estimate alone could understate expected logistics or storage needs. Classification: row 26 was a false positive with estimated risk probability 0.885; unnecessary follow-up is less dangerous than missing risk, but it can consume limited field-officer time. Cluster: support status dominates the selected two-group pattern, so the grouping should not be read as farm performance.

Improvement with real data

After informed consent, collect representative outcomes across seasons; document measurement quality; add agronomically justified variables; check subgroup errors and probability calibration; validate prospectively; monitor drift; and keep an appeal and human-review path.

STUDENT CHECK: Kawaya Gueylord must approve or rewrite the three interpretations above in their own words before submission and be able to explain them without reading a script.